ANIRBAN CHOWDHARY

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CURRENT POSITION

Position	**PhD Student**, School of Astrophysics, Presidency University, Kolkata
PhD completion	Expected June 2027

EDUCATION

2022–present	**Ph.D. in Astrophysics** (expected June 2027). Advisor: Dr. Suchetana Chatterjee. Tentative thesis title: 'Large Scale Structures in the High Redshift Universe'. PhD coursework GPA: 9.8/10. Presidency University · Kolkata
2020	**M.Sc. in Physics** Presidency University · Kolkata
2018	**B.Sc. (Hons.) in Physics** Presidency University · Kolkata

RESEARCH INTERESTS

My research interest is at the interface of cosmology and galaxy evolution. I use cosmological hydrodynamical simulations to study the connection between active galactic nuclei (AGN), their host galaxies and their dark matter halos. My current focus is on understanding the AGN-halo connection and particularly how AGN occupy halos as a function of luminosity, redshift, black hole mass and feedback. I try to model this co-evolution with halo occupation distributions and conditional luminosity functions. I am also interested in the statistics beyond the two-point function, such as k-nearest-neighbour distributions, and in using them to build feedback sensitive models of AGN clustering for current and upcoming and on going galaxy surveys.

RESEARCH EXPERIENCE

1. **PhD research** (2022–present)
   Advisor: Dr. Suchetana Chatterjee, Presidency University.

   - Quasar halo occupation in simulations. Measured the halo occupation distribution of quasars in IllustrisTNG and SIMBA and its dependence on luminosity, redshift, black hole mass and feedback mode. Showed that the central occupation of luminous AGN is suppressed in massive halos, and that a standard HOD fit to clustering can bias the inferred host halo mass.
   - AGN conditional luminosity functions. Modelled the AGN conditional luminosity function in TNG300 for $0.1 < z < 3$, and developed a joint MCMC framework that fits the luminosity function and two-point correlation function together to recover the host halos of quasars.
   - Beyond two-point statistics (ongoing, with Prof. Arka Banerjee). Testing whether k-nearest-neighbour cumulative distribution functions recover the galaxy– and AGN–halo connection in TNG300 where projected two-point fits fail.

2. **Master's research** (2019–2020)
   Topic: 'A Direct Measurement of the Mean Occupation Function of Obscured Quasars'.
   Advisor: Dr. Suchetana Chatterjee, Presidency University.
   Used WISE-selected obscured quasars and dark matter halos traced by Planck clusters to measure the halo occupation of obscured quasars directly.

3. **Bachelor's research** (2017–2018)
   Topic: 'Standardization of Type Ia Supernova Light Curves'.
   Advisor: Dr. Saumyadip Samui, Presidency University.
   Developed a code to standardize archival Type Ia supernova light curves for distance calibration.

PUBLICATIONS

Refereed

1. **Chowdhary, A.** & Chatterjee, S. 2025, "Halo Occupation Distribution of Quasars: Dependence on Luminosity, Redshift, Black Hole Mass and Feedback Modes", ApJ, 992, 21.

Submitted

2. **Chowdhary, A.**, Chatterjee, S. & Das, D. 2026, “Extracting Host Dark Matter Halos of Quasars at Intermediate Redshifts: A Combined Clustering and Luminosity Function Based Approach”, submitted to MNRAS.

In preparation

3. Dutta, S. et al. (incl. **Chowdhary, A.**), “Mock AGN catalogues from IllustrisTNG: a comparison of FSPS and CIGALE spectral energy distributions”, in preparation (student project mentored by A. Chowdhary).

INVITED AND CONTRIBUTED PRESENTATIONS

Sep 2026	Contributed poster on “Conditional Luminosity Functions of Active Galactic Nuclei in Simulations" at Cosmic Connections Second Edition: AI/ML Across Cosmic Scales 2026, held at IMSc Chennai, India.
May 2026	Contributed poster on “Halo Occupation Distribution of Quasars: Dependence on Luminosity, Redshift, Black Hole Mass and Feedback Modes." at at the 44th meeting of Astronomical Society of India, IIT Guwahati, Guwahati, India.
Mar 2026	Contributed talk on “Halo Occupation Distribution of Quasars: Dependence on Luminosity, Redshift, Black Hole Mass and Feedback Modes." at Neighbourhood Meeting: Chaos in the Cosmos, held at SINP Kolkata, India.
Mar 2025	Contributed talk on “Halo Occupation Distribution of Quasars: Dependence on Luminosity, Redshift, Black Hole Mass and Feedback Modes" at Young Astronomer’s Meet 2025, held at TIFR Mumbai, India.
Feb 2024	Contributed poster on “Halo Occupation Distribution of Quasars : Redshift Evolution and Effects of AGN Feedback" at the 42nd meeting of Astronomical Society of India hosted jointly by IISc, ISRO, and JNP, Bengaluru, India.
Dec 2023	Contributed poster on “Halo Occupation Distribution of Quasars : Redshift Evolution." at the 10th International Conference on Gravitation and Cosmology: New Horizons and Singularities in Gravity, IIT Guwahati, Guwahati, India.
Mar 2023	Contributed poster on “Halo Occupation Distribution: Redshift Evolution and Effect of Feedback" at the 41st meeting of Astronomical Society of India, IIT Indore, Indore, India.

MENTORING

2024–present	Siddharth Dutta: mock AGN catalogues from IllustrisTNG, comparing FSPS and CIGALE spectral energy distribution models (Dutta et al., in preparation). Guiding through the simulation data, sample selection and the design of the comparison.
2022–2023	Dweepsa Das: student research project, leading to co-authorship on the AGN conditional luminosity function paper (Chowdhary, Chatterjee & Das, submitted to MNRAS).

COMPUTATIONAL SKILLS

Programming	Expert: Python. Familiar: C/C++, Fortran
Simulations	IllustrisTNG, SIMBA, CAMELS, FLAMINGO
Software	Expert: Corrfunc, Halotools. Familiar: FSPS, CIGALE, yt
Methods	Halo occupation and conditional luminosity function modelling, two-point and kNN-CDF clustering statistics, MCMC parameter inference
Systems	Linux, high-performance computing clusters

PUBLIC OUTREACH

Mar 2025	Conducted a public outreach quiz at the Young Astronomers' Meet 2025, Tata Institute of Fundamental Research, Mumbai.
Dec 2023	Resource person for a teachers' training workshop organized by the International Astronomical Union's Office of Astronomy for Education (IAU-OAE) at Presidency University, Kolkata.
Dec 2022	Volunteer public communicator for Presidency University at the National Space Science Exhibition organized by ISRO and IISER Kolkata.

AWARDS AND FELLOWSHIPS

2022	Junior Research Fellowship, University Grants Commission National Eligibility Test (UGC-NET)